

Tushar Mittal

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Summary

DevOps and Cloud Platform Engineer with 3+ years of experience designing and operating scalable, high-availability infrastructure on **AWS** and **Kubernetes**. Deep expertise in CI/CD pipeline engineering (Jenkins, GitHub Actions, Argo CD), Infrastructure as Code (Terraform, Crossplane), and Python-based operational automation at scale. Hands-on with observability tooling while actively applying AI agents to operational workflows for automated alert analysis and incident diagnosis. Certified in **Kubernetes (CKAD, KCNA)** and **Terraform**. Active open source contributor serving as **Kubernetes Release Docs Lead for v1.37**.

Experience

Cloud Platform Engineer — Engineer I, FICO (Fair Isaac Corporation) June 2024 – present

- **Cut production incident MTTR by 60%** (45 to 18 min) by embedding kagent into the Prometheus/Alertmanager and LGTM stack to auto-surface root cause analysis and debug commands for **100+ weekly alerts**, eliminating manual log correlation for on-call engineers
- **Reduced CI/CD pipeline job startup latency by 80%** by optimizing GitHub self-hosted runners on Kubernetes (EKS) using EC2 Warm Pools and cluster-local image pre-fetching, cutting EC2 startup time by **75% (4 to 1 min)** and image fetch time by **~83% (60 to 10 sec)**, improving developer velocity across large-scale build environments
- **Increased CI/CD runner capacity 2.5x (270 to 700 pods)** with zero additional infrastructure cost by re-engineering AWS VPC CNI IP allocation and networking configuration across **8 Kubernetes (EKS) environments**, reclaiming **72%** wasted IPs
- **Developed a Golang-based Kubernetes Operator for Crossplane** wrapping the Imperva WAF Terraform provider, enabling declarative infrastructure as code with **100%** automated drift reconciliation and enforcing security standards across cloud platform environments
- **Saved 1300+ manual hours annually** and prevented SCIM license over-provisioning by developing Python automation that syncs Okta groups to Atlassian (Jira, Confluence), Backstage, and other platforms, reducing access provisioning time by **95%**

Cloud Platform Engineer — Intern, FICO (Fair Isaac Corporation) May 2023 – June 2024

- **Led zero-downtime migration of 400TB JFrog Artifactory** from shared to dedicated VPC by analyzing inbound flow logs across **100+** AWS accounts, developing a custom Python network topology mapper to reconstruct network paths, validate DNS and load balancer connectivity across environments
- **Reduced policy-related deployment failures by 70%** by building a Python CI/CD validation pipeline that maps Crossplane resource dependencies, renders infrastructure as code changes, and validates them against Kyverno policies in pull request checks

Awards

- **Extraordinary Impact Award** (top 5% company-wide) & **13 Spot Awards** at FICO (Fair Isaac Corporation)

Open Source and Community

Kubernetes (SIG Release)

- Release Team Contributor: **Docs lead** for v1.37 release cycle and **Docs shadow** for v1.35 and v1.36 release cycles
- Contributed to the official Kubernetes release tooling — [automated release note regeneration](#) and [CI pipeline enhancements](#)

PyCon India 2025 — Speaker: Making Sense of Big Codebases (DeepWiki & Crabviz)

Certifications

- **Kubernetes:** [CKAD](#), [KCNA](#) | **Terraform:** [Associate](#) | **GitHub:** [Foundations](#)

Technical Skills

Programming: Python, Go, Bash

Cloud Platforms: AWS (EC2, EKS, ECS, S3, RDS, VPC, ALB/NLB, IAM, KMS, SQS, SNS, EventBridge, CloudWatch, ECR)

Containers & Infrastructure: Kubernetes, Kubebuilder, Docker, Helm, Crossplane, Terraform, Ansible, Istio, Linux

CI/CD & Automation: Jenkins, GitHub Actions, Argo CD, GitLab CI, SonarQube, Artifactory

Observability & Policy: Prometheus, Grafana, Splunk, OpenTelemetry, Kyverno

Education

Chitkara University, BE in Computer Science | CGPA: 9.87/10.00 (Top 5%)

May 2024